

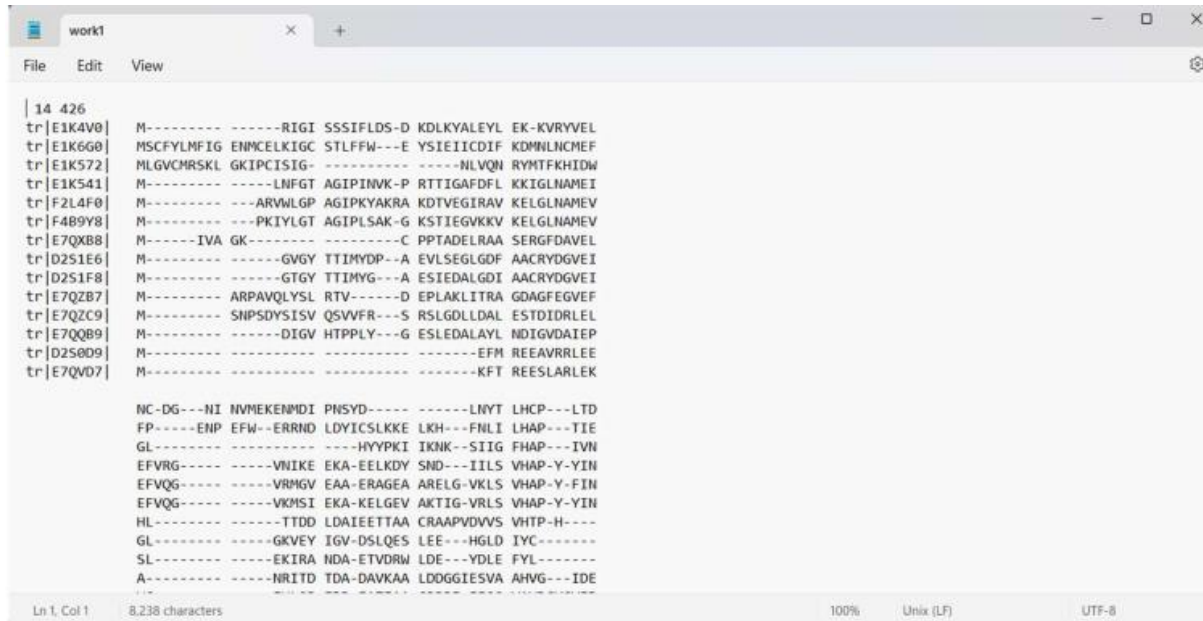
Bioinformatics

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Question 1

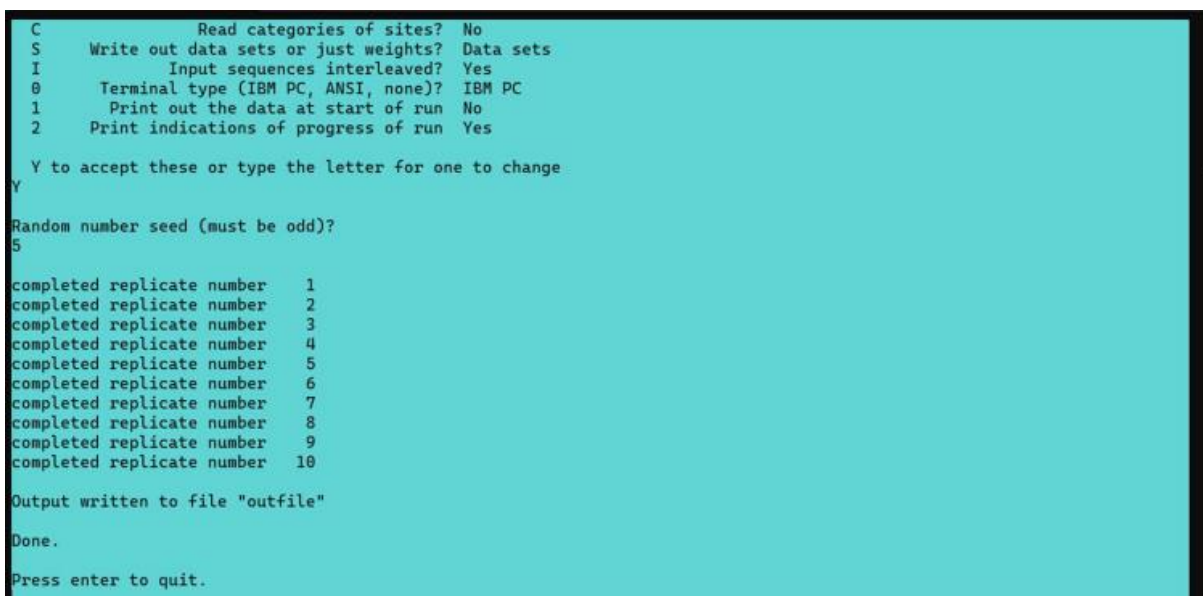
The initial sequences were processed through the MAFFT web server to perform multiple sequence alignment (MSA). The resulting alignment was saved in PHYLIP format within the directory containing the PHYLIP executable files.



```
| 14 426
tr|E1K4V0| M-----RIGI SSSIFLDS-D KDLKYALEYL EK-KVRYVEL
tr|E1K6G0| MSCFYLMFIG ENMCELKIGC STLFFW---E YSIEIICDIF KDMNLNCFE
tr|E1K572| MLGVCMRSLK GKIPICISIG- -----NLVQN RYMTFKHIDW
tr|E1K541| M-----LNFGT AGIPINVK-P RTTIGAFDFL KKIGLNAMEI
tr|F2L4F0| M-----ARVNLGP AGIPKYAKRA KDTVEGIRAV KELGLNAMEV
tr|F4B9Y8| M-----PKIYLG T AGIPLSAK-G KSTIEGVKKV KELGLNAMEV
tr|E7QXB8| M-----IVA GK-----C PPTADELRRA SERGFDAVEL
tr|D2S1E6| M-----GVGY TTIMYDP--A EVLSEGLGDF AACRYDGV EI
tr|D2S1F8| M-----GTGY TTIMYG--A ESIEDALGDI AACRYDGV EI
tr|E7QZB7| M-----ARPAVQLYSL RTV-----D EPLAKLITRA GDAGFEGVEF
tr|E7QZC9| M-----SHPSDYISIV QSVVFR--S RSLGDL L DAL ESTDIDRLEL
tr|E7QOB9| M-----DIGV HTPPLY--G ESLEDALAYL NDIGVDATIEP
tr|D2S009| M-----EFM REEAVRRLEE
tr|E7QV07| M-----KFT REESLARLEK

NC-DG---NI NVMEKENMDI PHSYD-----LNYT LHCP---LTD
FP----ENP EFW--ERRND LDYICSLKKE LKH---FNLI LHAP---TIE
GL-----HYPKI IKNK--SIIG FHAP---IVN
EFVRG-----VNIKE EKA-EELKDY SND---IILS VHAP-Y-YIN
EFVQG-----VRMGV EAA-ERAGEA ARELG-VKLS VHAP-Y-FIN
EFVQG-----VKMSI EKA-KELGEV AKTIG-VRLS VHAP-Y-YIN
HL-----TTDD LDAIEETAA CRAAPVDVVS VHTP-H---
GL-----GKVEY IGV-DSLQES LEE---HGLD IYC-----
SL-----EKIRA NDA-ETVDRW LDE---YDLE FYL-----
A-----NRITD TDA-DAVKAA LDDGGIESVA AHVG---IDE
```

Next, bootstrapping was carried out using PHYLIP's seqboot program.



```
C      Read categories of sites? No
S      Write out data sets or just weights? Data sets
I      Input sequences interleaved? Yes
0      Terminal type (IBM PC, ANSI, none)? IBM PC
1      Print out the data at start of run No
2      Print indications of progress of run Yes

Y to accept these or type the letter for one to change
Y
Random number seed (must be odd)?
5

completed replicate number 1
completed replicate number 2
completed replicate number 3
completed replicate number 4
completed replicate number 5
completed replicate number 6
completed replicate number 7
completed replicate number 8
completed replicate number 9
completed replicate number 10

Output written to file "outfile"
Done.
Press enter to quit.
```

The output file produced through this procedure serves as input for the proml program, which employs maximum likelihood to compute the phylogenetic tree.

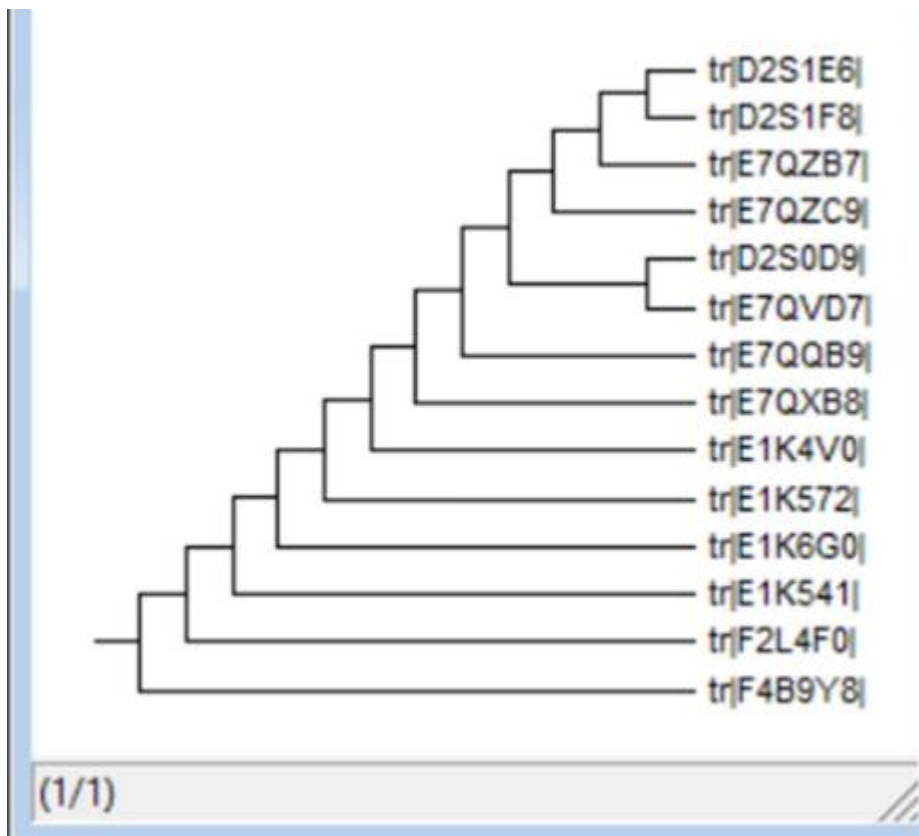
```

Amino acid sequence Maximum Likelihood method, version 3.695
Settings for this run:
Amino acid sequence Maximum Likelihood method, version 3.695
Settings for this run:
U      Search for best tree?  Yes
P      JTT, PMB or PAM probability model?  Jones-Taylor-Thornton
C      One category of sites?  Yes
R      Rate variation among sites?  constant rate of change
W      Sites weighted?  No
S      Speedier but rougher analysis?  Yes
G      Global rearrangements?  No
J      Randomize input order of sequences?  Yes (seed =      5, 3 times)
O      Outgroup root?  No, use as outgroup species 1
M      Analyze multiple data sets?  Yes, 10 data sets
I      Input sequences interleaved?  Yes
0      Terminal type (IBM PC, ANSI, none)?  IBM PC
1      Print out the data at start of run  No
2      Print indications of progress of run  Yes
3      Print out tree  Yes
4      Write out trees onto tree file?  Yes
5      Reconstruct hypothetical sequences?  No

Y to accept these or type the letter for one to change
y
Data set # 1:
Adding species:

```

The output from proml is fed into the consensus program to generate the consensus tree. Finally, TreeviewX is employed to visualize the tree.



A similar tree is built using the protdist program.

```

Protein distance algorithm, version 3.698

Settings for this run:
P Use JTT, PMB, PAM, Kimura, categories model? Jones-Taylor-Thornton matrix
G Gamma distribution of rates among positions? No
C One category of substitution rates? Yes
W Use weights for positions? No
M Analyze multiple data sets? Yes, 10 data sets
I Input sequences interleaved? Yes
0 Terminal type (IBM PC, ANSI)? IBM PC
1 Print out the data at start of run No
2 Print indications of progress of run Yes

Are these settings correct? (type Y or the letter for one to change)
y

Data set # 1:

Computing distances:
tr|E1K4V0|
tr|E1K6G0| .
tr|E1K572| ..
tr|E1K541| ...
tr|F2L4F0| ....
tr|F4B9Y8| .....
tr|E7QXB8| .....
tr|D2S1E6| .....
tr|D2S1F8| .....
tr|E7QZB7| .....
tr|E7QZC9| .....

```

The output of protdist is then input into the neighbor program.

```

Neighbor-Joining/UPGMA method version 3.698

Settings for this run:
N Neighbor-joining or UPGMA tree? Neighbor-joining
O Outgroup root? No, use as outgroup species 1
L Lower-triangular data matrix? No
R Upper-triangular data matrix? No
S Subreplicates? No
J Randomize input order of species? Yes (random number seed = 5)
M Analyze multiple data sets? Yes, 10 sets
0 Terminal type (IBM PC, ANSI, none)? IBM PC
1 Print out the data at start of run No
2 Print indications of progress of run Yes
3 Print out tree Yes
4 Write out trees onto tree file? Yes

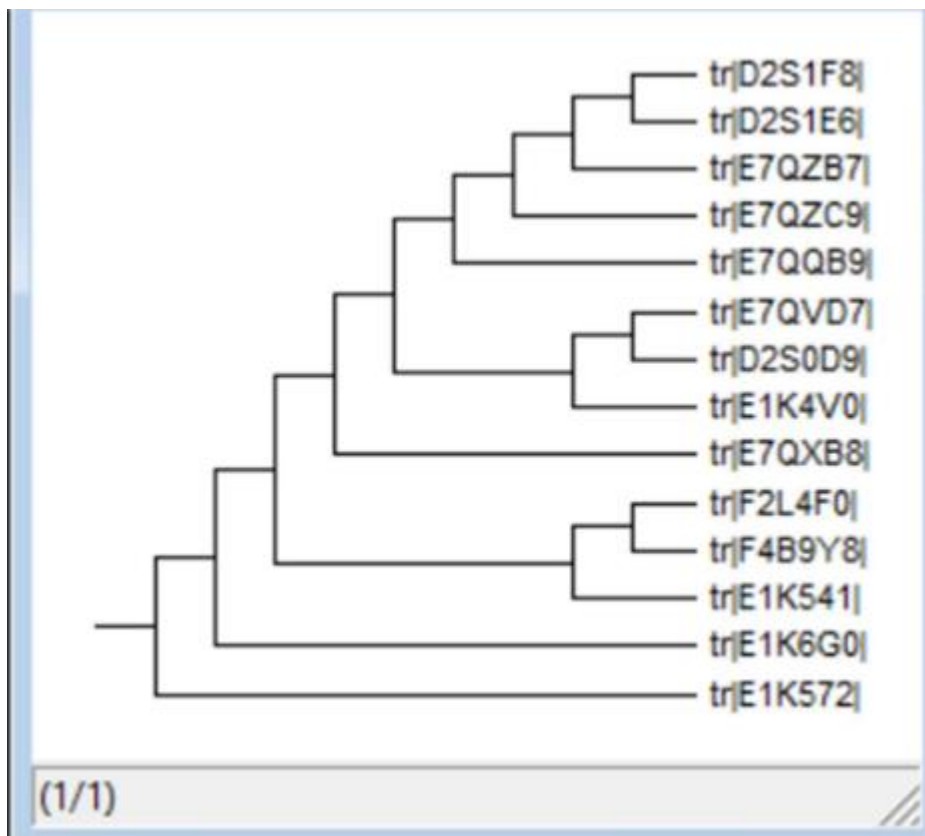
Y to accept these or type the letter for one to change
y

neighbor.exe: the file "outtree" that you wanted to
use as output tree file already exists.
Do you want to Replace it, Append to it,
write to a new File, or Quit?
(please type R, A, F, or Q)
f
Please enter a new file name> worklnj_tree
Data set # 1:

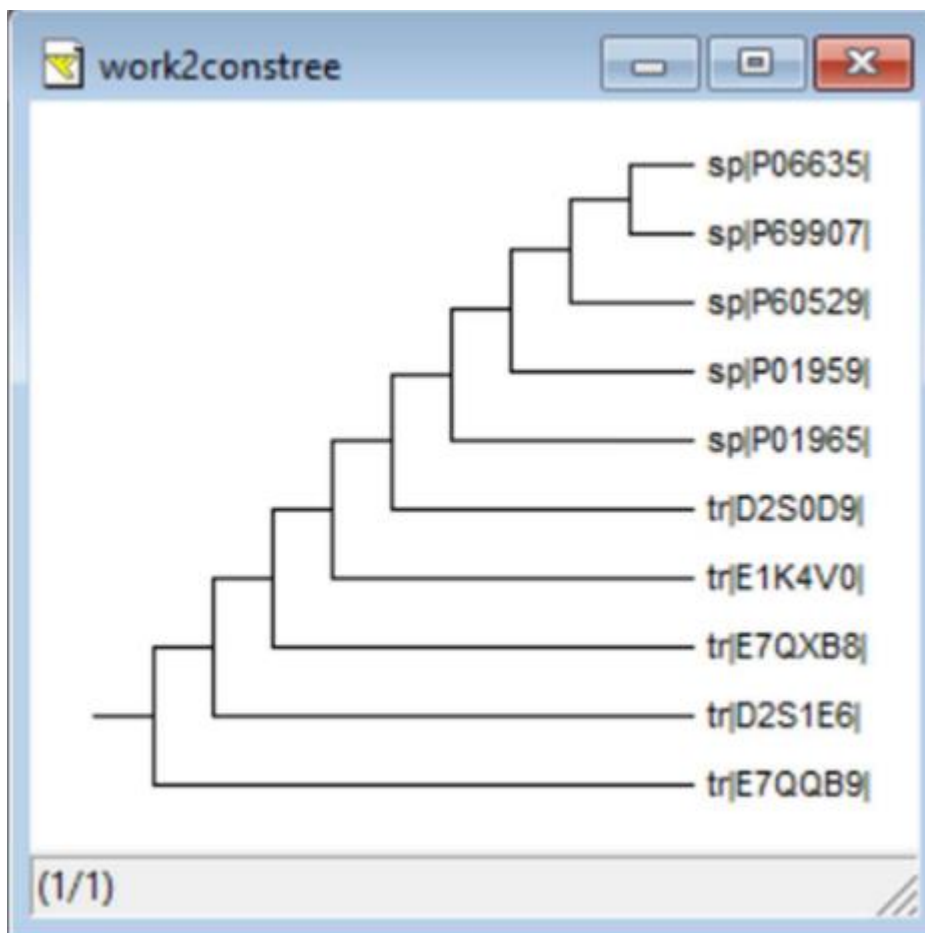
Cycle 11: species 14 ( 0.06312) joins species 13 ( 0.05480)
Cycle 10: species 5 ( 0.44123) joins species 6 ( 0.33488)
Cycle 9: species 9 ( 0.38905) joins species 8 ( 0.44665)

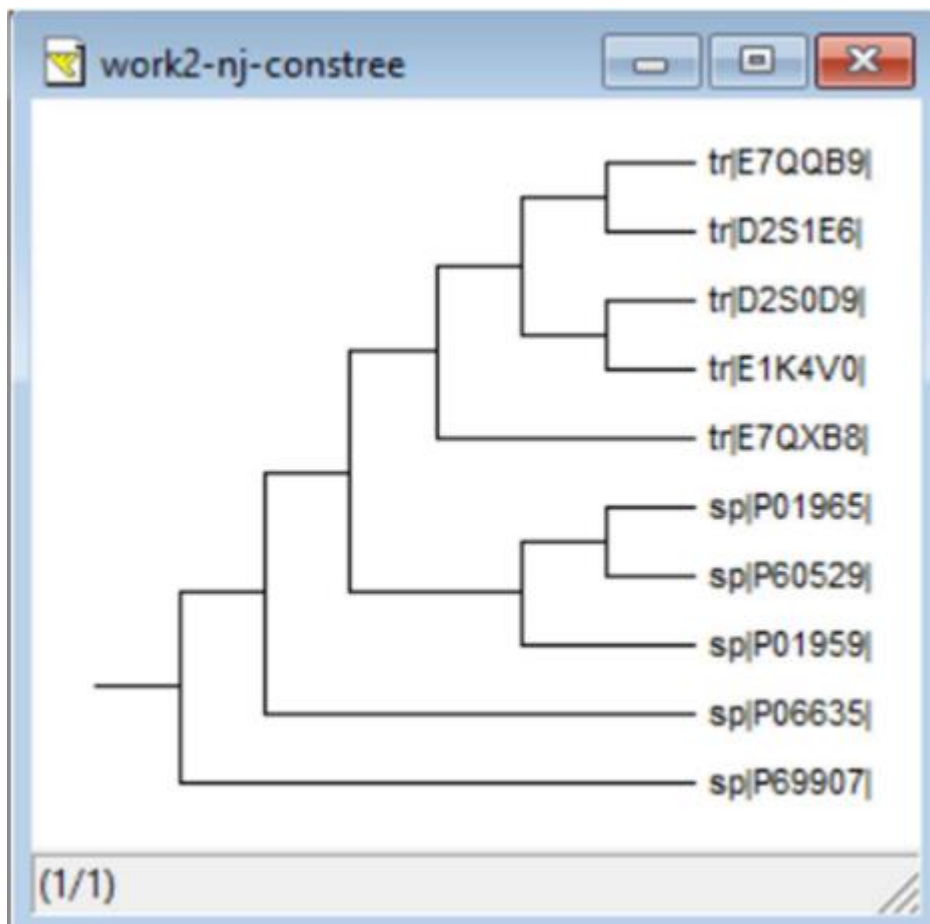
```

The consensus tree for this is generated using the consense program and visualized using TreeviewX.



The process is repeated for second set of sequences





Question 2

```
In [1]: import numpy as np
import pandas as pd

alignment = [
    "MVLSPADKTNVKGVGAHAGEYGAAAW",
    "MKRLPADPPCVKGVKAKAGDYGATTW",
    "MALSAADKTNVKSXVGGHAGEYGAATS",
    "MVLSAADKTNVKSXAGGNAGEWMAAAW",
    "MVLSAADKTNVKSXLANAGEFGAAAW",
    "ALLPVRTTYHKKASGHIPEKDLNNV",
    "DEASSLKGHHKKLEADALLIPLSASS"
]

residues = ["G", "A", "V", "L", "I", "P", "F", "Y", "W", "S", "T", "C", "M", "N", "Q", "D", "E", "K", "R", "H"]

alignment_df = pd.DataFrame([list(seq) for seq in alignment])
frequency_matrix = pd.DataFrame(index=residues, columns=range(1, len(alignment[0]) + 1))

for col in alignment_df:
    frequency_matrix[col + 1] = alignment_df[col].value_counts()

frequency_matrix.fillna(0, inplace=True)

N = len(alignment)
p = 1 / len(residues)

weight_matrix = frequency_matrix.apply(lambda x: np.log((x + p) / (p * (N + 1))).round(2))

print(weight_matrix.transpose())
```

	G	A	V	L	I	P	F	Y	W	S	T	C
1	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
2	-2.08	0.97	2.03	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
3	-2.08	0.97	-2.08	2.54	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
4	-2.08	-2.08	-2.08	0.97	-2.08	0.97	-2.08	-2.08	-2.08	2.54	-2.08	-2.08
5	-2.08	2.03	-2.08	-2.08	0.97	1.63	-2.08	-2.08	-2.08	0.97	-2.08	-2.08
6	-2.08	2.54	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
7	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08
8	0.97	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	0.97	-2.08
9	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	0.97	-2.08	-2.08	2.32	-2.08
10	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97
11	-2.08	-2.08	2.54	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
12	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
13	1.63	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	2.03	-2.08	0.97
14	-2.08	0.97	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
15	-2.08	0.97	2.32	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08
16	2.32	0.97	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
17	1.63	2.03	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
18	-2.08	0.97	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
19	-2.08	2.54	-2.08	0.97	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
20	2.54	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
21	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
22	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	0.97	2.03	0.97	-2.08	-2.08	-2.08
23	2.32	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	-2.08
24	-2.08	2.54	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08
25	-2.08	2.54	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08
26	-2.08	2.03	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	1.63	-2.08
27	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	2.32	1.63	-2.08	-2.08
	M	N	Q	D	E	K	R	H				
1	2.54	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08				
2	-2.08	-2.08	-2.08	-2.08	0.97	0.97	-2.08	-2.08				
3	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08				
4	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08				
5	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08				
6	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08				
7	-2.08	-2.08	-2.08	2.54	-2.08	0.97	-2.08	-2.08				
8	-2.08	-2.08	-2.08	-2.08	-2.08	2.32	-2.08	-2.08				
9	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97				
10	-2.08	2.32	-2.08	-2.08	-2.08	-2.08	-2.08	1.63				
11	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08				
12	-2.08	-2.08	-2.08	-2.08	-2.08	2.87	-2.08	-2.08				
13	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08				
14	-2.08	-2.08	-2.08	-2.08	-2.08	2.54	-2.08	-2.08				
15	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	-2.08				
16	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08				
17	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	0.97				
18	-2.08	1.63	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	1.63			
19	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08			
20	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08			
21	-2.08	-2.08	-2.08	0.97	2.54	-2.08	-2.08	-2.08	-2.08			
22	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	-2.08			
23	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08			
24	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08			
25	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08			
26	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08			
27	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08			